# Equation of a Circle

Exam code: 9MA0

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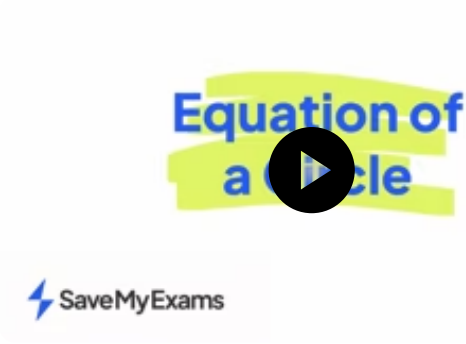


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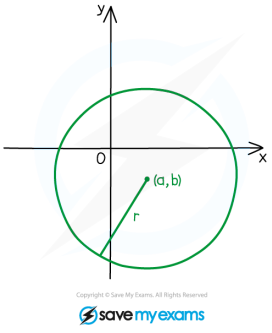
## Equation of a circle

### What is the equation of a circle?

- A circle with centre **(*a*, *b*)** and radius ***r*** has the equation

$$(x - a)^2 + (y - b)^2 = r^2$$




- You need to be able to find the centre and radius of a circle from its equation

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

e.g. $(x+4)^2+(y-7)^2=25$

THE CENTRE IS $(-4,7)$

THE RADIUS IS $r=\sqrt{25}=5$

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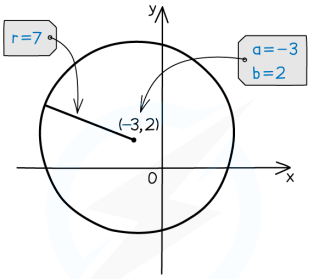
- You need to be able to find the equation of a circle given its centre and radius




THE EQUATION OF THE CIRCLE IS

$(x-(-3))^2+(y-2)^2=7^2$

$(x+3)^2+(y-2)^2=49$

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## Examiner Tips and Tricks

- Remember that the numbers in the brackets have the opposite signs to the coordinates of the centre.

$(x+12)^2+(y-9)^2=73$

CENTRE $(-12, 9)$

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- Don't forget to take the **square root** of the right-hand side of the equation when finding the radius.

## Worked Example

Find the equation of the circle.

THE RADIUS IS THE DISTANCE FROM $(-5,-7)$ TO $(1,1)$

$$r = \sqrt{(1-(-5))^2 + (1-(-7))^2}$$
$$= \sqrt{6^2+8^2} = \sqrt{100} = 10$$

THE DISTANCE BETWEEN TWO POINTS $(x_1, y_1)$ AND $(x_2, y_2)$ IS $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

THE EQUATION OF THE CIRCLE IS

$$(x-(-5))^2 + (y-(-7))^2 = 10^2$$
$$(x+5)^2 + (y+7)^2 = 100$$

$a = -5$
$b = -7$

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